

# PRAKHAR GARG

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## EDUCATION

|                                              |                                            |             |
|----------------------------------------------|--------------------------------------------|-------------|
| Delhi Technological University (DTU)         | B.Tech, Mathematics & Computing — CGPA 8.8 | 2022 - 2026 |
| Little Flowers Public Sr. Sec. School, Delhi | Class XII (CBSE) — Percentage 94.4%        | 2022        |
| Little Flowers Public Sr. Sec. School, Delhi | Class X (CBSE) — Percentage 97%            | 2020        |

## WORK EXPERIENCE

### Bain Capability Network | AI Engineer (Intern) January 2026 - Present

- Engineered an autonomous benchmarking agent that selects KPIs, computes sample sizes, and returns **structured outputs**, replacing a manual analyst workflow and improving **efficiency 20%**
- Architected multi-step AI agents and automated workflows with **LangChain**, **LangGraph**, and **LLM APIs**, handling tool selection, **API orchestration**, and graceful fallbacks with zero human intervention
- Built an end-to-end slides-to-answers **RAG system**, ingesting **1,000+ decks** via pipelines for **chunking**, **embedding**, **hybrid retrieval**, and **re-ranking**, significantly reducing hallucinations
- Designed production monitoring and feedback loops — **evaluation hooks** (groundedness, relevance), **latency/cost tracking**, **async I/O**, **caching**, and **retry/fallback**, improving AI reliability at scale
- Fine-tuned** and optimized **LLM prompts** and pipelines, applying **token cost reduction**, **caching**, and **model routing** to cut inference costs **~35%** while boosting response **accuracy**

### Indian Oil Corporation Limited (IOCL) | Gen AI (Intern) June – July 2025

- Rapidly built a document-summarization **RAG system** to query and synthesize internal documents, speeding up insight extraction from large unstructured datasets
- Implemented **data-preprocessing and chunking pipelines** for long-form text; applied prompt engineering and output evaluation to improve coherence, relevance, and factual consistency
- Developed **FastAPI backend services** with vector-based retrieval, structured logging, and guardrails

### PowerGrid | SDE (Intern) May – July 2024

- Built a full-stack file-sharing web app (**React**, **Node.js**, **Express**, **MongoDB**) on an **MVC architecture**, following software-engineering best practices — **Multer** for file handling, **Mongoose**, **CORS**

## PROJECTS

### Full-Stack MERN Chat App with Video Calling 2025

- Built and deployed a real-time chat & video-calling app (**MongoDB**, **Express**, **React**, **Node.js**) on **Render**; integrated the **Stream SDK** for scalable messaging/video with **JWT authentication**.
- Used **React Query** and **Zustand** for server-state management and dynamic theming (32 UI themes); modular **MVC backend** with a responsive **Tailwind + Daisy UI** front end.

## TECHNICAL SKILLS

**Language:** Python, Java, C++, C, SQL

**AI / GenAI:** RAG (hybrid retrieval, re-ranking, metadata filters), Multi-Agent Pipelines, LangChain, LangGraph, LangSmith, Prompt Engineering, Embeddings, Vector Search, Function / Tool Calling, MCP, Token Optimization, GPT-4o / OpenAI / Anthropic API

**Vector & Data:** FAISS, Pinecone, Chunking Strategies, MongoDB, PostgreSQL, MySQL

**Backend & APIs:** FastAPI, Node.js, Express.js, REST APIs, Microservices, JWT, Webhooks, Mongoose, CORS

**Frontend & Tools:** React.js, Vite, Tailwind CSS, Git, GitHub, Postman, Docker, AWS, Render, VS Code

## ACHIEVEMENTS

- Knight on LeetCode** (rating 2011), **Expert on Codeforces**; solved 500+ DSA problems across platforms
- Hackathons:** 2nd place at Hackster 2025; active participant in Brainwave 2.0, Hack & Chill 2.0, and Vihaan
- Chess: 1900 rating**, won 10+ inter-college tournaments
- CBSE All-India **Mathematics Topper** (Class 10 & 12) and All-India **Chemistry Topper** (Class 12)

## POSITIONS OF RESPONSIBILITY

### Cosmology Club – Web Development & Media Head

- Led a team of 12 students** to design, develop, and deploy the club's official website, overseeing media operations, coordinating content, design, and technical execution to boost the club's digital presence and event outreach

### Chess Club – Captain

- Captained the university chess team, leading players across inter-college tournaments and driving strategy and team selections